ZhongLi Shen (沈仲黎)

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Profiles

lelelemonade lelelemonade ZhongLi Shen

Experience

Software Engineer Intern (TP-LINK) **Jul 2020 - Dec 2020**

Backend Software Engineer (TP-LINK) **Dec 2020 - Jul 2023**

Senior Software Engineer (TP-LINK) **Jul 2023 - Apr 2024**

Education

HuaZhong University of Science and Technology **Apr 2016-Jun 2020**
Information Management and Information System bachelor

Projects

2FA For TP-LINK ID **2021.04-2021.08**

- Introduced 2FA feature for the 50 million users. Supporting email, push notification and OTP authentication methods.
- Designed extensible and unified external interfaces, RPC calls, and database schemas to ensure scalability.

Java, Spring Boot, MySQL, RabbitMQ, Cassandra, Redis, Prometheus, Grafana

SMS Service Refactoring **2022.03-2022.06**

- Refactored the legacy SMS service from scratch using Spring Boot.
- Support ctyun SMS, resulting in a daily cost reduction of 2000 Euro.
- Introduced automated CI/CD of SMS audit and publish, circuit breaker and retry mechanisms, improving the daily SMS success rate from 98% to 99.9%.

Spring Boot, Tencent Cloud, Aliyun, CTYun, RabbitMQ, MySQL, Prometheus, Grafana

Upgrade MySQL 5.6 to 5.7 **2022.01-2022.03**

- Upgraded the PRD MySQL RDS from version 5.6 to 5.7 to comply with AWS's mandatory upgrade plan.
- Evaluated the upgrade impact of MySQL 5.7, assessed the upgrade plan and tested compatibility with our applications in the PRE-PRD.
- Successfully completed the upgrade with an expected impact on write operations of expected 5 minutes.

AWS, SQL, MySQL, RDS, Golang

MySQL Spring Boot Starter **2023.01-2023.04**

- A mysql-spring-boot-starter using Druid and ShardingSphere. Allowed developers connect to MySQL through configuration.
- Supported circuit breaking, rate limiting, dynamic dataSource routing, read/write separation. Complemented existing monitoring, sensitive management, configuration hot-reloading and DDD library.
- Encapsulated with few bug, clear documentation and easy integration.

MySQL, Druid, Spring Boot, ShardingSphere

Skills

Java	Golang
Kotlin	Spring Boot
AWS	PostgreSQL
MySQL	Cassandra
Redis	Kubernetes
Docker	Kafka
RabbitMQ	Restful
gRPC	GraphQL
Prometheus	Grafana
Kibana	ShardingSphere
Junit	Github Action
JavaScript	React
Python	Rust
Linux	Agile

Certifications

Kotlin for Java Developer

By Jetbrains

Languages

English
fluent in speak/read/write/listen
●●●●○

Chinese
mother tongue
●●●●●

Work Eligibility

Citizenship: China

VISA Type: Job Seeking VISA
Qualified for EU Blue card

Projects

Distributed Task Scheduler 2023.04-2023.07

- Researched, designed, reviewed, and developed a company-level distributed job scheduler.
- Designed a single-threaded, lock-free scheduling model for single-node scenarios to reduce resource consumption. Used commit logs and periodic flushing to ensure persistence of scheduling progress.
- In cluster scenarios, improved scheduling performance using time wheel. Enhanced database read performance using distributed locks and segmented scanning, enabling horizontal scalability and high availability.
- Developed the task scheduler in Go language and provided SDKs for both Go and Java language executors, along with comprehensive documentation.

TimeWheel, Goroutine, Spring Boot, Zookeeper, PostgreSQL, Golang, gRPC

Cross-Region Invocation for In-House RPC 2022.12-2023.03

- Implemented the capability of cross-region invocation for our in-house RPC system with minimal modifications, resulting in significant reduction in cross-region bandwidth consumption (5 to 10 times).
- Identify bottleneck and migrated cross-region HTTP requests from OkHttpClient to WebFlux, reducing CPU usage and thread overhead in high-latency scenarios.
- The deployment of the updated cross-region services went smoothly, ensuring stable operation. It is estimated that the monthly cost of cross-region bandwidth will be reduced by 2000 Euro.
- The compressed cross-region request model is currently undergoing patent review at the **Patent Office**.

WebFlux, OkHttp, GraphQL, RPC, Spring Boot, gRPC, Java

Upgrade Cassandra 2.x to 3.x 2021.09-2021.12

- Tested the read and write performance of Cassandra 3.x under different garbage collectors and consistency levels. Completed detailed test reports and provided optimal configuration recommendations for PRD.
- Designed, tested online migration plan and developed data synchronization and data validation tools to ensure data integrity and accuracy.
- After successfully completing the online migration, we achieved a significant reduction in CPU usage (by 60%) and ensured that the maximum garbage collection pause time did not exceed 50 milliseconds.

Cassandra, Kafka, CQL, CAP, Golang

Sharding Migration for MySQL 2023.10-2023.12

- Performed sharding migration of a MySQL table with 60 million records, ensuring smooth transition with minimal impact for PRD. Designed, executed a reliable sharding strategy and migration plan.
- Reduced IO pressure on the MySQL database by 75%.

ShardingSphere, Golang, MySQL, Stress Test, Data Migration, Transaction, canal, binlog

Secret Management Using Vault 2022.07-2022.10

- Introduced Vault for secrets management. Designed processes for auditing, releasing, and managing secrets and infrastructure passwords. Supported multiple versions and environments.
- Developed spring-boot-starters that enabling seamless integrate into micro-services with minimal configuration changes.
- Developers were able to develop, build, test, and deploy without direct access to secrets.

Vault, CICD, Secret Management, Spring Boot, Kubernetes, Docker, Vault Agent